



CSE 4205

Digital Logic Design

Introduction

Course Teacher: Md. Hamjajul Ashmafee

Assistant Professor, CSE, IUT

Email: ashmafee@iut-dhaka.edu



Administration

- **Course Credit:** 3 (Total marks 300)
- **Lectures:** Tuesday and Wednesday
- **Sessional Works:** Thursday – Lab 6
- **Discussion:** Based on prior appointment



Administration

- **Textbooks:**

- Digital Logic and Computer Design – M. Morris Mano - 4E
- Digital Fundamental – Thomas L. Floyd – 8E

- **Grading:**

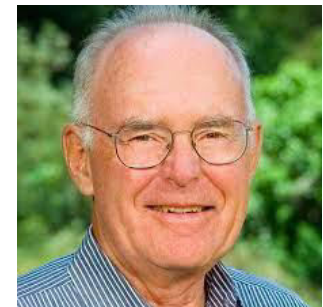
- Quiz/Assignment/Class performance: 15%
- Mid term examination: 25%
- Final examination: 50%
- Attendance: 10%

- **LMS:**

- Google Classroom, Zoom/ Google meet

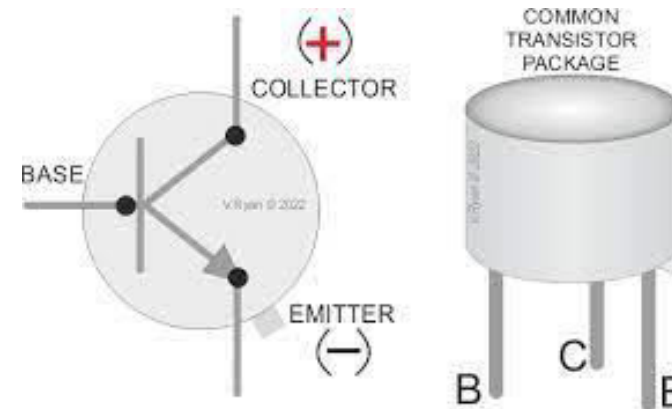
Motivation

- Revolution of **Microprocessor** and **Semiconductor** (Transistor)
- **Robert Noyce** (1927-1990)
 - “Mayor of Silicon Valley”
 - Confounder of Semiconductor
 - Confounder of INTEL
 - Co-inventor of IC
- **Gordon Moore**
 - Confounder of INTEL
 - **Moore’s Law**: the number of transistors on a computer chip **doubles** every year (observed in 1965)
 - Since 1975, transistor counts have doubled every two years.

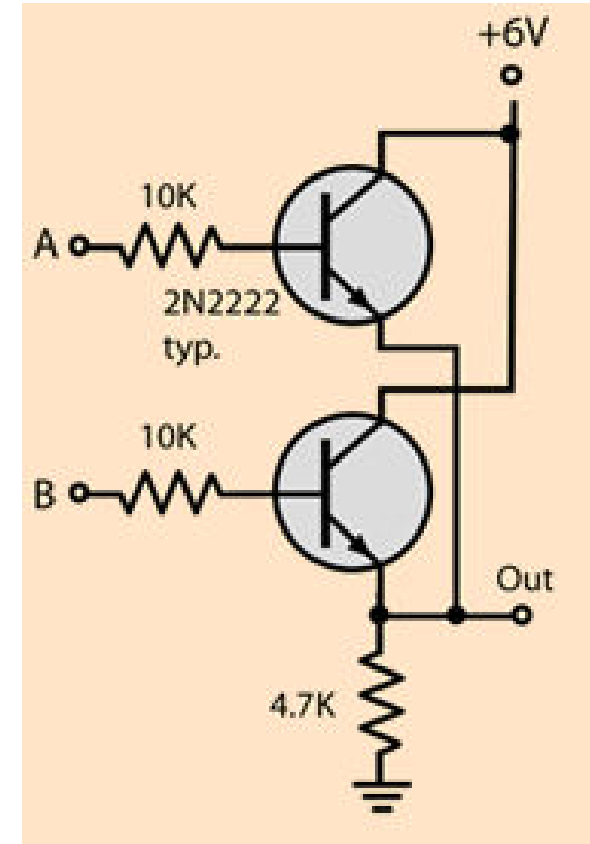
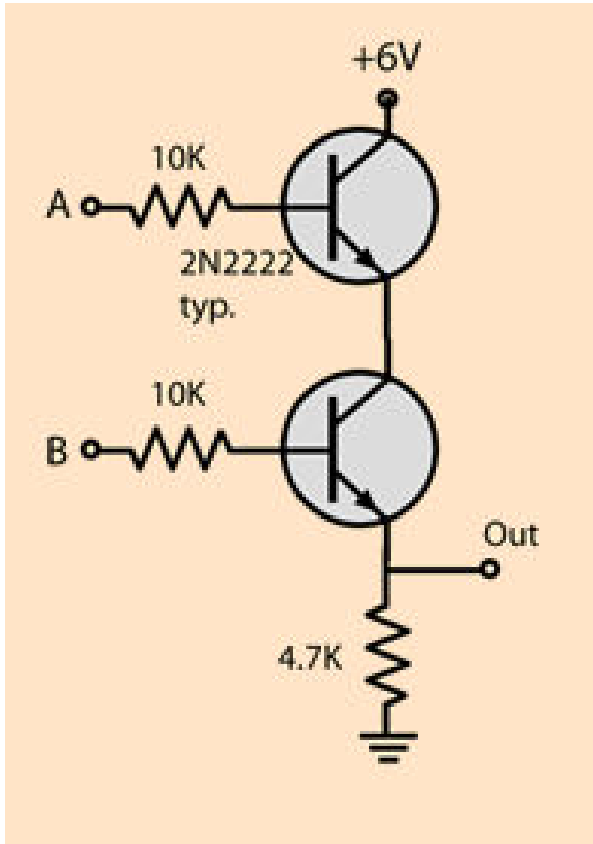


Semiconductor and Transistor

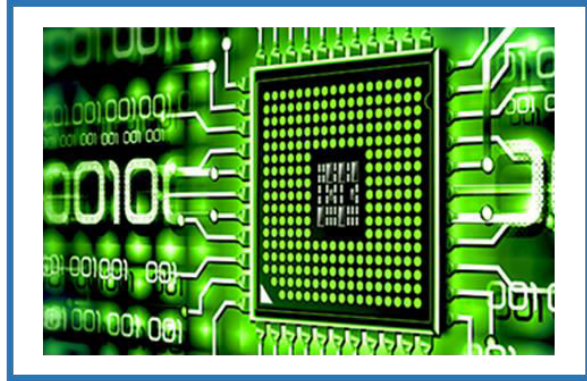
- A transistor is a semiconductor device used to **amplify or switch** electrical signals and power.
 - One of the basic **building blocks** of modern electronics



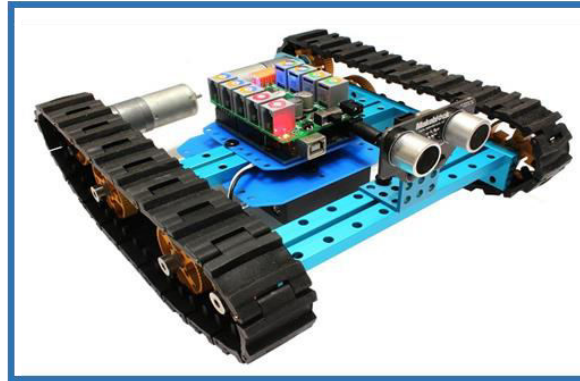
RTL diagram – AND and OR gate



Prospect of this Course



Computer Architecture



Robotics



Industrial Automation



Intelligent Security Control

... many more

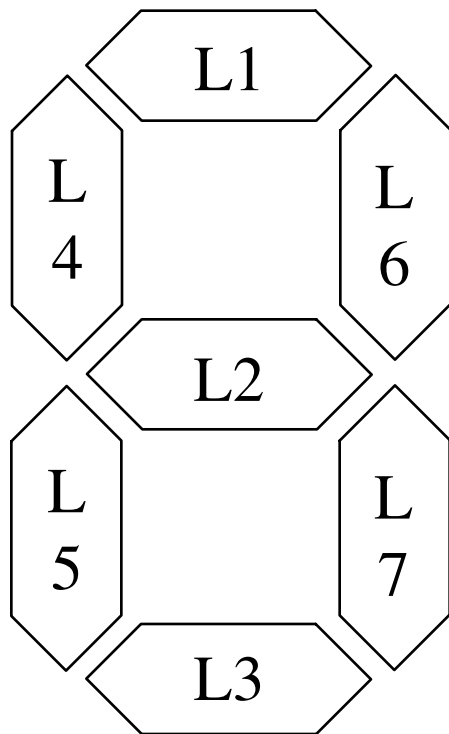


Prospect of this Course

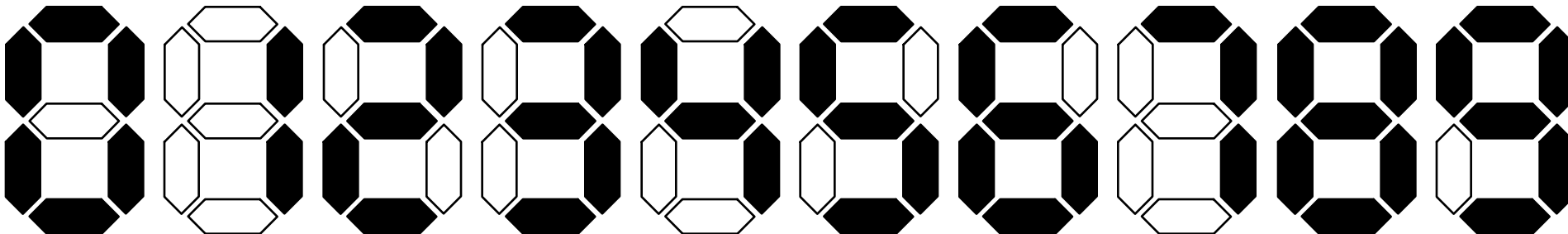
A good start is needed.

This course is very fundamental for the students who want to continue job/research in Computer design, Robotics etc. Simulation software for designing different logic gates will also be exercised. Students can form study group to solve the problems.

Case Study - 1



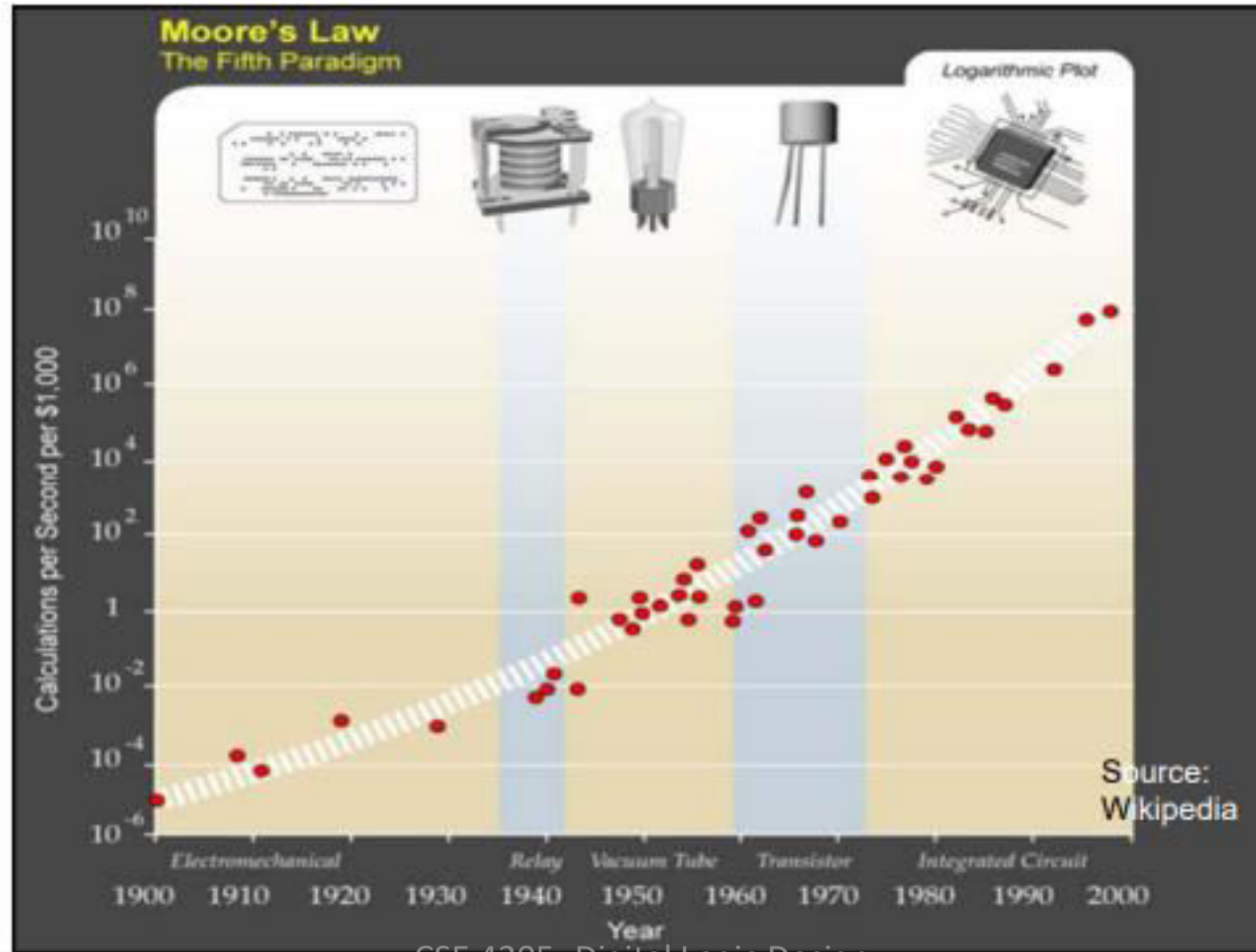
B3	B2	B1	B0	Val	L1	L2	L3	L4	L5	L6	L7
0	0	0	0	0	1	0	1	1	1	1	1
0	0	0	1	1	0	0	0	0	0	1	1
0	0	1	0	2	1	1	1	0	1	1	0
0	0	1	1	3	1	1	1	0	0	1	1
0	1	0	0	4	0	1	0	1	0	1	1
0	1	0	1	5	1	1	1	1	0	0	1
0	1	1	0	6	1	1	1	1	1	0	1
0	1	1	1	7	1	0	0	0	0	1	1
1	0	0	0	8	1	1	1	1	1	1	1
1	0	0	1	9	1	1	1	1	0	1	1



Case Study - 2

- Add two 1-bit binary numbers
 - **Input:** 1 bit augend and 1 bit addend
 - **Output:** 1 bit carry and 1 bit sum

Evolution of Computer





Basic Idea

- Logic gate [basic relationship] to build complex relationship
 - AND, OR, NOT
 - Definition
 - Symbol
 - Signals in waveform as INPUT
- Truth Table to formulate the relationship
- Logic expression: another way to re-express
 - Algebra and circuit
- Prepackaged Building block – IC
- Implementation of logic gates – Transistors
- Signals
 - Analog – turning on (switch on) lights
 - Digital – Calculator
 - In real-life a threshold value is considered to convert an analog system to a digital one

Circuit

- Relationship among signals
 - Where the signals are provided in waveforms in time
- Types:
 - Analog
 - Continuous values
 - Low accuracy and susceptible to noise
 - Digital
 - Discrete values
 - High accuracy and less susceptible to noise



23:59:59





ABC

- ABC



ABC

- ABC



ABC

- ABC



ABC

- ABC



ABC

- ABC